Validating a Bitcoin Forecasting Model: Look-Ahead Bias in VMD-Based Features

Unpredictable returns, predictable risk: risk cone and HAR volatility model

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Abstract

A multi-horizon CNN-BiLSTM model forecasting hourly Bitcoin returns from a Variational Mode Decomposition (VMD) of past returns appeared to predict the direction of the next hours correctly about 8 times out of 10. We show that this apparent skill is entirely due to look-ahead bias: when the VMD is applied to a whole block of data, each mode contains information from future prices. Once the decomposition is restricted to past data only, the same network, trained on the same data, does not outperform a random walk over a 15-month out-of-sample period. A Temporal Fusion Transformer trained on the same data does no better, confirming that the absence of predictive power does not depend on the architecture, and simple ARIMA benchmarks only find a tiny, economically unusable 1-hour effect. Risk, by contrast, is forecastable: a risk cone built from EWMA volatility with empirical fat-tailed multipliers passes the Kupiec coverage test, whereas the Gaussian cone is rejected, and a log-HAR model (Corsi, 2009) on realised variance from 5-minute returns forecasts next-day volatility significantly better than RiskMetrics EWMA.

Interactive application: https://btc-model-validation.streamlit.app/
Code: https://github.com/ldenarbonne/btc-forecast-model-validation

1 Model and data

Data. Hourly BTC/USDT closing prices from Binance, January 2018 to October 2026 (about 77,000 observations), with hourly log-returns $r_t = \ln P_t - \ln P_{t-1}$.

Features. The VMD splits a signal into $K = 3$ band-limited modes (penalty $\alpha = 2000$). The network receives the last 24 hourly values of the three modes.

Targets. Log-returns between successive horizons $h \in \{1, 2, 4, 6, 8, 12, 24\}$ hours, chained to obtain a forecast path.

Network. Two 1-D convolution layers (32 and 64 filters) followed by a two-layer bidirectional LSTM (64 units) and a linear output layer; Adam optimiser, mean squared error, early stopping on the validation set.

Chronological split. Training: January 2018 – June 2024; validation (early stopping): July 2024 – June 2025; test: July 2025 – October 2026 (11,240 hourly forecasts), never used for training or model selection. A 24-hour gap separates the periods so that targets do not overlap.

2 Two ways of computing the features

- **VMD on whole blocks** (original approach): the decomposition is run once on a long series (here, one calendar year at a time). The value of a mode at hour t then depends on the returns observed *after* t.

- **Causal VMD** (corrected approach): for each forecast time t, the decomposition is run only on the 256 returns ending at t. We checked that the features of a sample are strictly identical when all data after t is deleted.

3 Out-of-sample results

Accuracy is compared with a random walk (forecast of zero return), using the out-of-sample $R^2 = 1 - \sum(y - \hat{y})^2 / \sum y^2$ on cumulative log-returns, and the share of forecasts with the correct sign.

	Causal VMD		VMD on whole blocks	
Horizon	R^2 vs random walk	Direction correct	R^2 vs random walk	Direction correct
1h	-0.0005	50.3%	+0.62	75.8%
2h	-0.0021	50.6%	+0.71	79.5%
4h	-0.0021	49.9%	+0.62	76.5%
6h	-0.0023	49.3%	+0.42	71.4%
8h	-0.0020	50.6%	+0.21	66.6%
12h	-0.0007	51.1%	+0.00	58.6%
24h	-0.0006	51.5%	-0.05	56.2%

Table 1: Test period July 2025 – October 2026. Same network, data and split; only the feature computation differs.

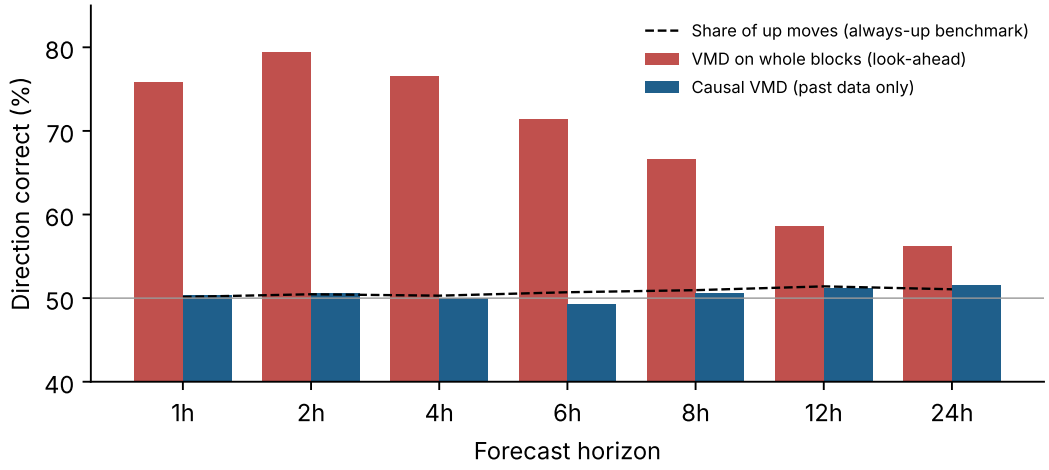


Figure 1: Share of correct directional forecasts. The dashed line is the share of up moves, i.e. the score of a naive "always up" forecast.

With look-ahead, the model appears to explain more than 60% of the variance of 1- to 4-hour returns and to forecast their direction correctly in 76–80% of cases. With the causal decomposition, the R^2 is slightly negative at every horizon and the directional accuracy (49–51.5%) is no better than the naive "always up" benchmark.

4 Comparison with a Temporal Fusion Transformer

To check whether a more advanced architecture changes the conclusion, a Temporal Fusion Transformer (TFT, implementation `pytorch-forecasting`) was trained on the same data, split

and test period. Its inputs are causal: the last 72 hourly log-returns, absolute returns and 24-hour rolling volatility, plus hour of day and day of week (known in advance). It forecasts the next 24 hourly returns (median of a quantile forecast).

Both models are evaluated on the same 11,240 forecast origins. Statistical significance is assessed with a Diebold–Mariano test against the random walk (Newey–West variance, one-sided p-value for "model more accurate") and a Pesaran–Timmermann directional test computed on non-overlapping samples, since overlapping multi-hour forecasts are not independent. The economic test is a long/short strategy on non-overlapping periods, net of 0.1% per trade.

	R^2 vs random walk		Direction correct		DM p-value		Annualised Sharpe (net)		
Horizon	CNN-BiLSTM	TFT	CNN-BiLSTM	TFT	CNN-BiLSTM	TFT	CNN-BiLSTM	TFT	Buy & hold
1h	-0.0005	-0.0011	50.3%	50.1%	0.66	0.89	-3.5	-9.4	-0.5
2h	-0.0021	-0.0021	50.6%	50.2%	0.83	0.94	-2.5	-2.5	-0.5
4h	-0.0021	-0.0043	49.9%	50.1%	0.84	0.96	-3.5	-0.7	-0.5
6h	-0.0023	-0.0062	49.3%	50.5%	0.81	0.97	-2.5	-0.9	-0.5
8h	-0.0020	-0.0080	50.6%	51.0%	0.75	0.96	-3.3	-0.7	-0.5
12h	-0.0007	-0.0111	51.1%	51.1%	0.57	0.96	-1.4	-0.6	-0.5
24h	-0.0006	-0.0252	51.5%	50.2%	0.57	0.98	-0.8	-0.9	-0.5

Table 2: Out-of-sample comparison, July 2025 – October 2026.

Neither model beats the random walk: out-of-sample R^2 is negative at every horizon, no Diebold–Mariano test is significant, and the Pesaran–Timmermann p-values on non-overlapping samples lie between 0.24 and 0.86. The TFT is slightly worse at longer horizons: it learned the strong upward drift of 2018–2024 and forecasts a rise in 97% of cases at 24 hours, which is costly over a test period in which Bitcoin fell. This illustrates regime dependence: a model extrapolates the drift of its training period. The TFT used here is deliberately compact (about 18,000 parameters) and trained with a limited computing budget.

5 Simple time-series benchmarks

Classical models were estimated on the training period and applied with fixed parameters to the same 11,240 test origins: an ARIMA(2,1,0), whose order is selected by BIC, an ARIMA(1,1,1), a seasonal SARIMA$(2,1,0)(2,0,0)_{24}$ capturing the daily cycle, and a constant drift. Table 3 reports the 1-hour horizon, the only one at which any model shows a detectable signal. Besides the R^2, we report the skill of the mean absolute error (MAE) relative to the random walk, with a Diebold–Mariano test on absolute errors, and the Gini coefficient (Accuracy Ratio, AR $= 2\,\mathrm{AUC} - 1$) of the forecast as a score for the sign of the next return, with a block-bootstrap 95% confidence interval.

Model (1-hour horizon)	R^2 vs RW	Direction	MAE skill (DM p)	Gini [95% CI]	Sharpe (net)
CNN-BiLSTM + VMD	-0.0005	50.3%	-0.00% (0.53)	2.6% [0.7; 4.7]	-3.5
TFT	-0.0011	50.1%	-0.09% (0.97)	0.0% [-2.0; 2.0]	-9.4
ARIMA(2,1,0)	-0.0015	52.6%	+0.21% (0.0004)	6.4% [4.6; 7.9]	-13.5
ARIMA(1,1,1)	-0.0009	52.7%	+0.18% (0.00002)	6.4% [5.0; 7.9]	-13.7
SARIMA$(2,1,0)(2,0,0)_{24}$	-0.0033	51.8%	+0.08% (0.17)	4.2% [2.5; 6.3]	-17.4
Constant drift	-0.0001	50.1%	–	–	-0.5

Table 3: Test period July 2025 – October 2026. RW: random walk. Sharpe: annualised, long/short strategy net of 0.1% per trade.

The ARIMA models capture a small negative autocorrelation of hourly returns (bid–ask bounce and short-term mean reversion): about 52.6% correct directions, a Gini of 6%, and an MAE 0.2% lower than the random walk, statistically significant. The effect is, however, far too

small to be exploited: once transaction costs are included, the strategy loses heavily, because it trades every hour. At horizons of 2 hours and more, no model is distinguishable from the random walk. Simple models do at least as well as deep networks, which is consistent with the absence of exploitable signal.

6 Forecasting risk: an empirical risk cone

Since the expected return is essentially zero, the useful information in a forecast lies in its dispersion. The application therefore displays a risk cone around the current price. Hourly volatility σ_t is estimated by EWMA (RiskMetrics, $\lambda = 0.94$) on hourly log-returns, using only data available at t. For each horizon h (1 to 24 hours) and confidence level q, the band is

$$P_t \exp\Big(\pm\, c(h,q)\, \sigma_t \sqrt{h}\Big)\,,$$

where the multiplier $c(h,q)$ is the empirical q-quantile of $|\ln P_{t+h} - \ln P_t|/(\sigma_t\sqrt{h})$ over the calibration period (2018 – June 2025). The Gaussian cone uses $c = \Phi^{-1}\Big(\frac{1+q}{2}\Big)$ instead (1.96 at 95%, 2.58 at 99%). The cone is backtested on the test period with the Kupiec (1995) unconditional coverage test, computed on non-overlapping observations. Scenarios shown in the application are simulated by bootstrapping standardised hourly returns of the calibration period, scaled by the current EWMA volatility.

	95% band				99% band			
	Empirical		Gaussian		Empirical		Gaussian	
Horizon	Coverage	p	Coverage	p	Coverage	p	Coverage	p
1h	94.9%	0.67	93.3%	0.00	99.1%	0.16	96.6%	0.00
2h	94.8%	0.38	92.8%	0.00	99.2%	0.20	96.3%	0.00
4h	95.1%	0.38	92.2%	0.00	99.2%	0.10	95.7%	0.00
6h	95.2%	0.29	91.6%	0.00	99.2%	0.68	95.5%	0.00
8h	95.6%	0.04	91.3%	0.00	99.3%	0.15	95.7%	0.00
12h	96.0%	0.37	91.3%	0.00	99.4%	0.64	95.8%	0.00
24h	96.2%	0.10	90.3%	0.00	99.7%	0.16	95.0%	0.00

Table 4: Coverage of the risk cone on the test period (July 2025 – October 2026) and Kupiec p-values. Empirical multipliers at 95%: 2.19 (1h) to 2.79 (24h); at 99%: 4.18 to 4.94.

The Gaussian cone is rejected at every horizon and level ($p < 0.001$): at 99%, it misses 3 to 5% of outcomes instead of 1%, a direct consequence of the fat tails of Bitcoin returns (the empirical 99% multiplier is 4.2 to 4.9 standard deviations instead of 2.58). The empirical cone is not rejected at the 5% level in 13 of 14 cases; the only rejection (8 hours, 95%, $p = 0.04$) corresponds to a slight *over*-coverage, i.e. a conservative band.

7 Forecasting volatility: the HAR model

Realised variance. Daily realised variance is computed from 5-minute log-returns, $RV_t = \sum_{i=1}^{288} r_{t,i}^2$ (Bitcoin trades 24/7), over 2,840 days from 2018 to October 2026.

Model. The heterogeneous autoregressive model of Corsi (2009) explains future realised variance by its daily, weekly and monthly averages, reflecting traders with different horizons. Since the crypto market trades every day, the weekly and monthly averages use 7 and 30 days. We use the log version, which is more robust to the spikes of realised variance:

$$\ln RV_{t+1:t+h} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln RV_t^{(7)} + \beta_m \ln RV_t^{(30)} + \varepsilon_{t+h},$$

where $RV_{t+1:t+h}$ is the average realised variance over the next h days ($h = 1$ and 7). The forecast is $\exp(\hat{\beta}' x_t + \hat{\sigma}^2_\varepsilon/2)$ (lognormal bias correction). Parameters are estimated by OLS on January 2019 – June 2024 and kept fixed. For $h = 1$: $\hat{\beta}_d = 0.45$, $\hat{\beta}_w = 0.33$, $\hat{\beta}_m = 0.12$; their sum (0.91) measures the strong persistence of volatility.

Benchmarks and loss functions. The log-HAR is compared with the HAR in levels, the random walk on realised variance, EWMA RiskMetrics ($\lambda = 0.94$) on daily squared returns, EWMA on realised variance, and a GARCH(1,1) with Student-t innovations. The main loss is QLIKE, $L = RV/\hat{\sigma}^2 - \ln(RV/\hat{\sigma}^2) - 1$, which is robust to the measurement noise of realised variance (Patton, 2011). We also report the RMSE of annualised volatility, the R^2 of the Mincer–Zarnowitz regression of realised on forecast variance, and a one-sided Diebold–Mariano test on QLIKE against EWMA RiskMetrics (Newey–West variance).

	Next day (474 days)				Next 7 days, average (468 days)			
Model	QLIKE	RMSE (vol pts)	MZ R^2	DM p	QLIKE	RMSE (vol pts)	MZ R^2	DM p
log-HAR	**0.272**	**15.7**	**0.21**	**0.004**	**0.196**	14.9	**0.21**	0.24
HAR-RV (levels)	0.430	21.6	0.18	0.95	0.360	22.6	0.18	0.94
EWMA RiskMetrics (daily returns)	0.342	16.8	0.11	–	0.233	13.9	0.12	–
EWMA on realised variance	0.364	17.1	0.09	0.93	0.225	**13.2**	0.12	0.36
GARCH(1,1) Student-t	0.384	20.2	0.11	0.83	0.236	17.5	0.13	0.52
Random walk (realised variance)	0.451	17.4	0.15	1.00	0.230	14.2	0.14	0.45

Table 5: Out-of-sample volatility forecasts, July 2025 – October 2026. RMSE in points of annualised volatility. DM p: one-sided test of lower QLIKE than EWMA RiskMetrics.

At the 1-day horizon, the log-HAR reduces the QLIKE loss by about 20% relative to RiskMetrics, and the difference is significant ($p = 0.004$); it also explains twice as much of the variation of realised variance (Mincer–Zarnowitz R^2 of 0.21 against 0.11). The HAR in levels is penalised by the spikes of realised variance, which distort OLS estimation: the log transformation matters as much as the model itself. At 7 days, the log-HAR still has the lowest QLIKE, but the gain is no longer significant, as the forecasts of all models converge towards the average level of volatility.

8 Conclusions

1. Signal decompositions (VMD, EMD, wavelets) applied to a full series are *non-causal*: they must be recomputed on a rolling window of past data, both in training and in production.

2. A directional accuracy far above 50% on hourly crypto returns should be treated as a red flag for leakage rather than as evidence of skill.

3. Out-of-sample tests must be chronological, with a gap between periods, and compared with a simple benchmark (random walk, always-up).

4. Model complexity is no substitute for signal: a Temporal Fusion Transformer trained on the same data does no better than the CNN-BiLSTM or a random walk.

5. In this case, the corrected model has no forecasting power: hourly Bitcoin returns behave essentially like a random walk at these horizons. Simple ARIMA models detect a tiny 1-hour mean reversion, statistically significant but wiped out by transaction costs.

6. Risk, unlike direction, is forecastable. Bands must be calibrated on the empirical distribution: Gaussian quantiles underestimate tail risk and are rejected by the Kupiec test.

7. For volatility, a simple and transparent model, the log-HAR on realised variance, beats RiskMetrics EWMA and GARCH out of sample at the 1-day horizon.

8. The value of the exercise lies in the validation framework (causal features, chronological split, benchmarks, statistical tests, backtesting), which is directly transferable to model risk management and to the validation of market risk models.

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